```
\documentclass[11pt,a4paper]{article}

\usepackage[T1]{fontenc}
\usepackage[utf8]{inputenc}
\usepackage{amsmath,amssymb,amsthm}
\usepackage{booktabs}
\usepackage{longtable}
\usepackage{geometry}
\usepackage{hyperref}
\usepackage[expansion=false]{microtype}
\usepackage{siunitx}
\usepackage{algorithm}
\usepackage{algpseudocode}

\geometry{margin=2.5cm}
\hypersetup{colorlinks=true,linkcolor=black,citecolor=black,urlcolor=blue}
\sisetup{group-separator={,},group-minimum-digits=4}

\newcommand{\pn}{p(n)}
\newcommand{\Cemp}{C_{\mathrm{EMP}}}
\newcommand{\li}{\operatorname{li}}
\newtheorem{observation}{Observation}
\newtheorem{remark}{Remark}
\theoremstyle{definition}
\newtheorem{definition}{Definition}

\title{Empirically Calibrated Intervals for the $n$-th Prime:\\
Construction, Measurement, and Limits}

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\small Independent Researcher, Egypt}

\date{\today}

\begin{document}
\maketitle

\begin{abstract}
We describe in full a computational procedure that, given the primes up
to $N$, produces an interval containing the $n$-th prime $\pn$ for any
$n \in (N,2N]$. Across eight scales from $n=10^{8}$ to
$n=6\times10^{15}$ the interval contained the true value in every case,
and was $107$ to $3{,}608$ times narrower than the tightest proven bounds
\cite{axler2019}; an exhaustive test at smaller scales covered
```

$3{,}840{,}000$ consecutive targets with a 100% containment rate.

This paper documents the construction completely: every equation, the measured basis for every constant, the alternatives that were tried and rejected with the numbers that rejected them, and the questions that remain unanswered. We treat the negative results as part of the contribution, since several of them close directions that would otherwise look promising.

The central limitation is stated at the outset and repeated where it matters: \emph{the interval is empirical, not proven}. The calibration constant extrapolates a maximum-value statistic, and no theorem relates future maxima to past maxima. The only unconditional component is the Axler bound, applied as a clipping ceiling.
\end{abstract}

\noindent\textbf{Keywords:} n-th prime, prime counting function, Riemann R function, zeta zeros, empirical bounds, computational number theory\\
\textbf{MSC 2020:} 11N05, 11Y11, 11Y35

\tableofcontents

\section{Introduction}

\subsection{The two established approaches}

Two distinct traditions address the location of $\pn$.

\paragraph{Explicit bounds.}
Rosser and Schoenfeld \cite{rosser1962} gave the first effective estimates. Dusart \cite{dusart2018} and then Axler \cite{axler2019} tightened them successively. Axler's Theorem 1.1, valid for $n \ge \num{46254381}$, states
\begin{equation}\label{eq:axler-upper}
\pn < n\left(\ln n + \ln\ln n - 1 + \frac{\ln\ln n - 2}{\ln n}
- \frac{\ln^2\ln n - 6\ln\ln n + 10.667}{2\ln^2 n}\right),
\end{equation}
and Theorem 1.2, valid for $n \ge 2$, gives the matching lower bound with the constant 10.667 replaced by 11.508 and the inequality reversed.

These are unconditional. Their weakness is width: at $n=2\times10^{15}$ the interval spans about 1.3×10^{12}.

\paragraph{Exponentially tight effective bounds.}
A recent development in the same direction is due to Visser \cite{visser2025}, who derives bounds of the form

\begin{equation}\label{eq:visser}
\li^{-1}\!\left(n\left[1 - a(\ln[n\ln n])^{b+1}
  e^{-c\sqrt{\ln[n\ln n]}}\right]\right) < \pn <
\li^{-1}\!\left(n\left[1 + a(\ln[n\ln n])^{b+1}
  e^{-c\sqrt{\ln[n\ln n]}}\right]\right)
\end{equation}
from effective bounds on $|\pi(x)-\li(x)|$. These are structurally close to the present work: both localise $\pn$ by inverting a counting function rather than by expanding an asymptotic series. We compare against them in \S\ref{sec:visser}.

\paragraph{Exact computation.}
The Meissel--Lehmer family \cite{lmo1985,dr1996,gourdon2001}, implemented in \texttt{primecount} \cite{primecount}, computes $\pi(x)$ and hence $\pn$ exactly. On ordinary hardware $p(10^{12})$ returns in milliseconds.

\subsection{What this work is, and what it is not}

\emph{This is not a competitor to exact computation.} Where \texttt{primecount} applies it is faster and exact, and two of the four operating modes described below call it internally.

Nor is it a theorem. The interval produced carries no proof.

What it is: a direct measurement of how far the best available analytic estimates deviate from the truth, and an interval calibrated to that measurement. An exact algorithm returns a number with no uncertainty description. A proven bound describes uncertainty conservatively. This work measures the deviation and reports the interval that follows.

\subsection{Operating range}

Training on $[1,N]$ yields a valid containment on $(N,2N]$ and no further. Every result reported here is at a ratio of exactly 2.00.

This is not an incidental restriction. The calibration constant $\Cemp$ (\S\ref{sec:cemp}) is measured on the training range and applied beyond it; the further one extrapolates, the weaker the justification. The single-doubling choice keeps the extrapolation at a distance where it has been tested.

\subsection{How to read this paper}

Sections \ref{sec:invr}--\ref{sec:branching} build the construction, each component with its derivation and the measurement that justifies its form. Section \ref{sec:modes} describes the four modes. Section

```
\ref{sec:results} reports results. Sections
\ref{sec:negative}--\ref{sec:open} document what failed and what remains
unresolved; these are not appendices to the contribution but part of it.

Numbers reported as ``measured'' are from runs whose parameters are given
at the point of use. Where a measurement was made on an earlier or
modified version of the code, this is stated.

\section{Notation and conventions}

\begin{center}
\begin{tabular}{ll}
\toprule
Symbol & Meaning\\
\midrule
$\pn$ & the $n$-th prime\\
$\pi(x)$ & the number of primes $\le x$\\
$\li(x)$ & the logarithmic integral, $\int_0^x dt/\ln t$ (Cauchy p.v.)\\
$R(x)$ & the Riemann prime counting function, \eqref{eq:R}\\
$\mu(k)$ & the M\"obius function\\
$\gamma_k$ & the imaginary part of the $k$-th non-trivial zeta zero\\
$N$ & the training bound; the primes in $[1,N]$ are known\\
$F_b(n)$ & the analytic base estimator, \eqref{eq:Fb}\\
$F_z(n)$ & the full estimator including the zeta layer, \eqref{eq:Fz}\\
$r(n)$ & the residual $\pn - F_z(n)$\\
$\Cemp$ & the empirical deviation constant, \eqref{eq:cemp}\\
$[L_n,U_n]$ & the Axler interval at $n$\\
$[x_1,x_2]$ & the interval produced by the procedure\\
\bottomrule
\end{tabular}
\end{center}

Throughout, ``the model'' refers to the complete procedure, and
``residual'' to $r(n)$ unless qualified.

\section{The analytic base}\label{sec:invr}

\subsection{Why the Riemann function rather than the logarithmic integral}

The prime number theorem gives $\pi(x)\sim\li(x)$, and one might invert
$\li$ to estimate $\pn$. This is not the best available choice, and the
reason is structural rather than empirical.

Riemann's function
\begin{equation}\label{eq:R}
R(x) = \sum_{k\ge1}\frac{\mu(k)}{k}\li\!\left(x^{1/k}\right)
\end{equation}
```

arises from M\"obius inversion of
\begin{equation}\label{eq:J}
J(x) = \sum_{k\ge1}\frac{1}{k}\pi\!\left(x^{1/k}\right)
     = \pi(x) + \tfrac12\pi\!\left(x^{1/2}\right)
     + \tfrac13\pi\!\left(x^{1/3}\right) + \cdots,
\end{equation}
the function for which $\li$ is the natural approximation. In other words, $\li(x)$ approximates a count that includes prime \emph{powers}; $R(x)$ removes that excess and approximates $\pi(x)$ directly.

The excess is not negligible at the scales of interest. At $x = 10^{12}$, the term $\tfrac12\pi(x^{1/2}) \approx \num{39250}$, which is comparable to the entire deviation we are trying to bound.

\paragraph{Measured effect.}
Over 400 targets outside the training range, using each estimator as the base with no further correction:

\begin{center}
\begin{tabular}{lrr}
\toprule
Base & Mean $|$error$|$ & Ratio\\
\midrule
$\li^{-1}(n)$ & \num{47075} & ---\\
$R^{-1}(n)$ & \num{6381} & $7.4\times$\\
\bottomrule
\end{tabular}
\end{center}

A further measurement across scales shows the advantage is not constant: R^{-1} was between $1.4\times$ and $62\times$ closer than $\li^{-1}$ depending on the scale, the advantage being largest where the prime-power correction is proportionally largest.

\subsection{Computing R^{-1} at scale}

Direct evaluation of \eqref{eq:R} requires roughly seven evaluations of Ei (terms with $\mu(k)=0$ vanish; the series is truncated once $x^{1/k}<2$). Over the hundreds of millions of points required for a full $\Cemp$ scan (\S\ref{sec:cemp}) this is prohibitive.

We exploit the smoothness of the difference. Define
\begin{equation}\label{eq:D}
D(x) = R(x) - \li(x)
     = \sum_{k\ge2}\frac{\mu(k)}{k}\li\!\left(x^{1/k}\right).
\end{equation}
This is dominated by $-\tfrac12\li(x^{1/2})$, which is smooth and slowly

varying. We tabulate D once on a fixed logarithmic grid of
$\num{200000}$ points spanning $[2,10^{18}]$ and interpolate linearly.

\begin{observation}
The interpolation error of D on this grid is at most about
8×10^{-6}, against residuals of order 10^{3}--10^{4}. It is
therefore not a limiting factor.
\end{observation}

\begin{remark}[Why the grid range must be fixed]
An earlier implementation sized the grid to the current problem. This
made results depend on the run: the same n evaluated in two different
sessions differed by about 10^{-6}, which propagated into the
branching logic (\S\ref{sec:branching}) and produced different intervals.
The grid range is now a compile-time constant.
\end{remark}

Inversion is by Newton's method on $R(x)=n$, using $R'(x)=1/\ln x$:
\begin{equation}\label{eq:newton}
x_{j+1} = x_j - \left(R(x_j)-n\right)\ln x_j.
\end{equation}

The first step is free. Seeding with $x_0 = \li^{-1}(n)$ gives
$\li(x_0)=n$ exactly, hence $R(x_0)-n = D(x_0)$ by \eqref{eq:D}, so
\begin{equation}
x_1 = x_0 - D(x_0)\ln x_0
\end{equation}
requires no evaluation of Ei at all. A single further step
with the true $\li$ suffices:

\begin{algorithm}
\caption{$R^{-1}(n)$}
\begin{algorithmic}[1]
\State $x \gets \li^{-1}(n)$ \Comment{seed; Newton on $\li$}
\State $x \gets x - D(x)\ln x$ \Comment{free step, uses tabulated D}
\State $x \gets x - \left(\li(x)+D(x)-n\right)\ln x$
       \Comment{one Ei evaluation}
\State \Return x
\end{algorithmic}
\end{algorithm}

\subsection{A cheaper inverse for the calibration scan}

The $\Cemp$ scan (\S\ref{sec:cemp}) evaluates R^{-1} at every integer
up to N. Even one Ei call per point is costly there.

For that loop only, we replace $\li$ in the early Newton iterations by

its asymptotic series
$$\li(x) \approx \frac{x}{\ln x}\left(1 + \frac{1}{\ln x}
+ \frac{2}{\ln^2 x} + \frac{6}{\ln^3 x} + \cdots\right), \tag{eq:liasym}$$
which requires no special function, and apply a single correction with the true $\li$ at the end to remove the series bias.

\begin{remark}[The series fails at small argument]
Series \eqref{eq:liasym} diverges from $\li$ for small x: measured error reaches 89% at $n=10$. Points below a threshold are therefore computed by the exact route. Since they are few, the cost is negligible. With this guard, the maximum error of the fast inverse across a mixed range is about 10^{-12}, and it runs $2.2\times$ faster than the exact version on large inputs.
\end{remark}

The zeta-zero layer

Motivation from the explicit formula

Riemann's explicit formula expresses $J(x)$ in terms of the non-trivial zeros $\rho=\tfrac12+i\gamma$:
$$J(x) = \li(x) - \sum_{\rho}\li\!\left(x^{\rho}\right)
       - \ln 2 + \int_x^\infty\frac{dt}{t(t^2-1)\ln t}. \tag{eq:explicit}$$
Pairing conjugate zeros, each contributes an oscillation of the form
$$\li\!\left(x^{\rho}\right) + \li\!\left(x^{\bar\rho}\right)
\;\approx\; \frac{2\sqrt{x}}{\ln x}\cdot
\frac{\cos\!\left(\gamma\ln x - \arg\right)}{|\rho|}, \tag{eq:zeropair}$$
with amplitude decaying as $1/|\rho|\approx 1/\gamma$.

This motivates --- but does not derive --- our basis. We are estimating $\pn$, not $\pi(x)$, and we apply the basis at $R^{-1}(n)$ rather than at x. The construction below should be read as a regression basis inspired by \eqref{eq:zeropair}, not as a truncation of \eqref{eq:explicit}.

Construction

For each of the first $K=60$ zeros we form the pair
$$\begin{aligned} A_k(x) &= \frac{\sqrt{x}}{\gamma_k \ln x},\\
\varphi_{2k}(x) &= A_k(x)\cos(\gamma_k\ln x),\qquad
\varphi_{2k+1}(x) = A_k(x)\sin(\gamma_k\ln x),\end{aligned} \tag{eq:basis}$$

\end{align}
giving $2K = 120$ features. Writing $b = R^{-1}(n)$, the base estimator is
\begin{equation}\label{eq:Fb}
F_b(n) = b = R^{-1}(n),
\end{equation}
and the full estimator is
\begin{equation}\label{eq:Fz}
F_z(n) = F_b(n) + \sum_{j=0}^{2K-1}\beta_j\varphi_j(b) + c_0.
\end{equation}

The coefficients β are fitted by ridge regression on the residuals $\pn - F_b(n)$ over a grid of the training range. The penalty is chosen by cross-validation over a logarithmic sweep.

\subsection{The role of c_0}

c_0 is the median of $\pn - F_b(n) - \sum\beta_j\varphi_j$ over \emph{all} grid points, computed after the regression.

Its purpose is to centre the estimator on the known sample. It matters that it is a median rather than a mean (robustness to the extreme residuals documented in \S\ref{sec:clustering}), and that it is computed over all points rather than a subset --- a distinction that becomes important in \S\ref{sec:cremoved}.

\subsection{Measured effect}

Over 400 targets outside the training range:

\begin{center}
\begin{tabular}{lrr}
\toprule
Estimator & Mean $|$error$|$ & Ratio to $\li^{-1}$\\
\midrule
$\li^{-1}(n)$ & \num{47075} & ---\\
$F_b(n) = R^{-1}(n)$ & \num{6381} & $7.4\times$\\
$F_z(n)$, \eqref{eq:Fz} & \num{3574} & $13.2\times$\\
\bottomrule
\end{tabular}
\end{center}

The zeta layer contributes a factor of 1.79 over R^{-1} alone, and improves the estimate on 68.8% of the 400 targets.

\subsection{Negative result: additional zeros degrade the estimate}\label{sec:morezeros}

The natural question is whether more zeros help. We computed zeros 61--1060 (up to $\gamma = 1489$) and repeated the fit, evaluating out of sample with a temporal split so that no target informed its own fit.

\begin{center}
\begin{tabular}{rrr}
\toprule
Zeros used & $\max\gamma$ & Out-of-sample improvement\\
\midrule
60 (baseline) & 169 & ---\\
200 & 486 & $+4.99\%$\\
500 & 888 & $+2.90\%$\\
1{,}000 & 1{,}489 & $+1.19\%$\\
\bottomrule
\end{tabular}
\end{center}

The improvement \emph{decreases} monotonically as zeros are added. Two mechanisms combine:

\begin{enumerate}
\item By \eqref{eq:basis} the amplitude of the k-th feature scales as $1/\gamma_k$. The 1000th zero contributes at roughly 11% of the amplitude of the first, so its signal-to-noise ratio in a finite sample is correspondingly worse.
\item Ridge regression penalises all coefficients jointly. Adding 1880 low-amplitude features forces the cross-validated penalty upward, which suppresses the coefficients of the informative low zeros along with the uninformative high ones.
\end{enumerate}

We therefore retain $K=60$. We note this is a statement about this regression setup at these sample sizes, not about the explicit formula: in \eqref{eq:explicit} the high zeros do carry information, and a different estimator might extract it.

\section{The empirical bound}\label{sec:cemp}

\subsection{Definition}

\begin{definition}
For a training bound N and a lower cutoff n_0,
\begin{equation}\label{eq:cemp}
\Cemp = \max_{n_0 \le n \le N}
        \frac{\left|\pn - F_z(n)\right|}{\ln^2 n}.
\end{equation}

```
\end{definition}

Two choices in \eqref{eq:cemp} require justification: the normalisation
$\ln^2 n$ and the cutoff $n_0$. Both are addressed below, and neither has
a theoretical basis --- a fact we consider important enough to state
before giving the measurements that motivated them.

\subsection{The normalisation has no theoretical
justification}\label{sec:normjust}

The factor $\ln^2 n$ is borrowed from Cram\'er's conjecture
\cite{cramer1936}, which concerns the largest gap between consecutive
primes near $x$:
\begin{equation}
\limsup_{n\to\infty}\frac{p_{n+1}-p_n}{\ln^2 p_n} = 1
\quad\text{(conjectural)}.
\end{equation}

Two objections apply, and we state both:

\begin{enumerate}
\item Cram\'er's statement is a conjecture, not a theorem. Granville
  \cite{granville1995} has argued the constant is likely larger than $1$.
\item More seriously, it describes \emph{prime gaps}, whereas
  \eqref{eq:cemp} measures \emph{model residuals}. These are different
  quantities. A large residual can arise without a large gap, and
  conversely.
\end{enumerate}

The normalisation is therefore an empirical choice that happens to have a
suggestive form. Section \ref{sec:normalt} reports what happened when we
tested alternatives --- including one that is better motivated
theoretically and performed worse.

\subsection{The lower cutoff is critical}\label{sec:cutoff}

Measured at $N = 2\times10^{6}$:

\begin{center}
\begin{tabular}{rrl}
\toprule
Cutoff $n_0$ & $\Cemp$ & Location of the maximum\\
\midrule
10 & 64.250 & $n = 11$\\
100 & 16.800 & $n = 111$\\
1{,}000 & 9.9 & $n \approx 1{,}200$\\
\num{10000} & 7.553 & $n = \num{1887313}$\\
```

```
\num{100000} & 7.553 & $n = \num{1887313}$\\
\bottomrule
\end{tabular}
\end{center}

At $n=11$ we have $\ln^2 11 = 5.75$. The residual there is small in
absolute terms; the ratio is large only because the denominator is
trivially small. This is an artefact of the normalisation at small
argument, not an extremal event.

From $n_0 = 10^4$ upward the value is stable, and the maximum falls at
$n=\num{1887313}$ --- a location that corresponds to a genuine large
prime gap. We adopt $n_0=\num{10000}$.

\begin{remark}[Impact of the error]
An earlier version used $n_0=10$. This inflated $\Cemp$ by a factor of
$8.5$, and hence the interval by the same factor, at every scale. The
error was invisible in the containment rate --- a too-wide interval still
contains --- and was found only by printing the \emph{location} of the
maximum rather than its value.
\end{remark}

\subsection{Structure of the maximum}\label{sec:maxstruct}

$\Cemp$ is a maximum-value statistic. Its behaviour as $N$ grows
determines whether extrapolating it is defensible, so we measured that
behaviour directly.

Tracking the running maximum over $2\times10^{7}$ points:

\begin{center}
\begin{tabular}{lr}
\toprule
Quantity & Value\\
\midrule
Total updates to the running maximum & 451\\
Number of clusters (updates within \num{10000} of each other) & 20\\
Median inter-cluster distance & \num{207454}\\
Maximum inter-cluster distance & \num{4689667}\\
Coefficient of variation of inter-cluster distances & 1.46\\
\bottomrule
\end{tabular}
\end{center}

\begin{observation}\label{obs:clustered}
A coefficient of variation of $1.46$ exceeds the value $1.0$ expected of
a Poisson process. The updates are therefore \emph{more clustered than
```

```
random}: long quiet stretches, of millions of points with no change at
all, punctuated by bursts.
\end{observation}

Two trends accompany increasing scale:

\begin{center}
\begin{tabular}{lr}
\toprule
Correlation & $r$\\
\midrule
Position vs.\ distance to the next cluster & $+0.52$\\
Position vs.\ size of the increment & $-0.45$\\
\bottomrule
\end{tabular}
\end{center}

Clusters become rarer and increments smaller as $N$ grows. This is
encouraging for extrapolation, but Observation \ref{obs:clustered}
prevents it from being conclusive: a clustered process can be quiet for a
long time and then move sharply.

\begin{remark}[Consequence for the margin]
Observation \ref{obs:clustered} is, we believe, the reason no regular law
for the safety margin could be derived (\S\ref{sec:margin}). One cannot
fit a trend to a process whose increments arrive in rare bursts of
variable size.
\end{remark}



\section{The two models and containment by intersection}\label{sec:intersect}

\subsection{Why two models}

The estimator $F_z$ of \eqref{eq:Fz} gives a point. To obtain an interval
one needs a second source of information, and the natural one is the
inverse relation: while the $\pn$ model asks ``where is the $n$-th
prime?'', a $\pi(x)$ model asks ``how many primes are below $x$?''.

We build both. Each is a stochastic walker (\S\ref{sec:branching}) that
maintains a set of branches; at any query point each yields not a single
value but a set of admissible values --- what we call the \emph{cloud}.

\subsection{The intersection}

Write $\mathrm{cloud}_\pi(x) \subset \mathbb{N}$ for the set of counts
```

the π model admits at x. The interval is
\begin{align}\label{eq:intersect}
x_1 &= \inf\left\{x : n \in \mathrm{cloud}_\pi(x)\right\},\\
x_2 &= \sup\left\{x : n \in \mathrm{cloud}_\pi(x)\right\},\notag
\end{align}
subsequently clipped by the Axler bound:
\begin{equation}\label{eq:clip}
\left[\max(L_n, x_1),\ \min(U_n, x_2)\right].
\end{equation}

The clipping in \eqref{eq:clip} is what makes the reported interval never worse than the proven one. Where the calibrated interval is narrower --- which is the case at every scale reported --- the result is empirical; where it would be wider, the proven bound takes over.

\subsection{Why not a fixed envelope}\label{sec:envelope}

The obvious alternative is to centre an envelope on the point estimate:
\begin{equation}\label{eq:env}
\left[F_z(n) - m\,\Cemp\ln^2 n,\ F_z(n) + m\,\Cemp\ln^2 n\right]
\end{equation}
for some multiplier m. This is simpler, faster, and uses only one model. We implemented it and measured it against the intersection at scales where the two disagree:

\begin{center}
\begin{tabular}{lll}
\toprule
& Intersection \eqref{eq:intersect} & Envelope \eqref{eq:env}\\
\midrule
$N = 7\times10^{6}$ & 0 failures, min.\ margin 19.1%
                    & 1 failure, min.\ margin 0.53%\\
$N = 1.2\times10^{7}$ & 0 failures, min.\ margin 32.6%
                      & 4 failures, min.\ margin 0.19%\\
\bottomrule
\end{tabular}
\end{center}

The envelope fails. The reason is that \eqref{eq:env} is symmetric and uniform: it applies the same half-width regardless of local structure. The intersection is neither. It draws on a second, independent model, and the width it yields at a given n reflects how quickly cloud_π moves through n --- which varies with the local density of primes.

\begin{remark}[This decided a design question]
Before this measurement it was not clear that the second model earned its

cost; it roughly doubles the computation. The measurement settled it. We
record it because the envelope is the natural first thing to try, and it
looks adequate until tested at a scale where it is not.
\end{remark}

\subsection{Seed independence}\label{sec:seed}

The search in \eqref{eq:intersect} is implemented by expanding outward
from a starting point. Because cloud_π has gaps --- regions
where the admissible set excludes n --- a naive search can stop at a
gap and return a boundary that depends on where it started.

Measured: of four targets tested with different starting points, three
gave different bounds, differing by 1--3%.

The fix is to run the search from several starting points distributed
across the admissible region and take the union of the resulting
intervals. We use 5 seeds, placed at evenly spaced quantiles of the
admissible set; with this the result is stable across repeated runs. This is a co
requirement, not an optimisation: without it the reported interval is not
a well-defined function of n.

\section{The branching mechanism}\label{sec:branching}

\subsection{Structure}

Both models advance in steps. The $\pn$ model steps in n with spacing
$\mathrm{CP}_{pn}=40$; the π model steps in x with spacing
$\mathrm{CP}_{\pi}\ln x$ where $\mathrm{CP}_\pi=5$.

Each branch carries an offset o, a slope correction s, and two
counters. At each checkpoint the observed local slope
\begin{equation}
\widehat{f}' = \frac{dF_b}{dn} + s + \frac{dW}{dn}
\end{equation}
is compared against a reference slope t, and the branch takes one of
four actions:

\begin{center}
\begin{tabular}{ll}
\toprule
Condition & Action\\
\midrule
$\left|\widehat{f}'-t\right|/|t| \le \theta$ & \textsc{hold}: keep o\\
value above the upper threshold & \textsc{brake}: decrease o\\
value below the lower threshold & \textsc{jump}: increase o\\
otherwise & split into three branches\\

\bottomrule
\end{tabular}
\end{center}

Splitting is capped: when the branch count exceeds MX the set is pruned to SEL by taking an evenly spaced subset in offset order, which preserves the extremes (which determine the interval) while bounding cost.

\subsection{Thresholds}\label{sec:thresh}

The upper and lower thresholds that separate \textsc{brake} and \textsc{jump} from splitting are set at $\pm T$ around the base, with
\begin{equation}\label{eq:thresh}
T = 2.28 \cdot N^{0.486} \cdot 0.84.
\end{equation}

The exponent 0.486 was obtained by fitting the observed scale of residuals against N over the training range; the leading constant and the factor 0.84 were set during development and not subsequently optimised. We report \eqref{eq:thresh} as it stands rather than as a derived quantity: it is a fitted form with two hand-set constants, and we have no argument that it is the right one.

The tolerance θ in the \textsc{hold} test starts at 0.02 and adapts with branch history. Writing c for the number of consecutive brakes or jumps a branch has taken and h for its consecutive holds:
\begin{align}\label{eq:adapt}
\theta &\gets \min\left(0.05,\ 0.02 \cdot 1.5^{\,\min(c,5)}\right)
  &&\text{if } c \ge 3,\\
\theta &\gets \max\left(0.005,\ \theta \cdot 0.85^{\,h}\right)
  &&\text{if } h \ge 5,\notag
\end{align}
and the correction factor applied when the branch does act starts at 0.60 and decays as $\max\left(0.30,\ 0.60\cdot0.9^{\,c}\right)$.

The intent is that a branch which has been correcting repeatedly is given room to continue, while one that has been stable is held to a tighter tolerance. The constants in \eqref{eq:adapt} are design choices; we did not test alternatives.

\subsection{The reference slope}

The reference t for the $\pn$ model derives from the lower convex hull of the points $\left(n,\pn\right)$ over the training range. Specifically, the hull is computed by the monotone chain algorithm, its edge slopes are extracted, and t is taken at the first index where consecutive

integer parts of the slope differ by at least 2.

For the π model, $t = 1/\ln x$ directly, this being the derivative of $\li$.

\begin{remark}[Memory in the hull computation]
The hull is over N points. A direct implementation materialising n and $\pn$ as float arrays and pre-allocating the hull required about 20\,GB at $N=5\times10^{8}$ --- an actual crash. The current implementation generates n internally, passes the primes in their stored integer form, and grows the hull dynamically. The result is bit-identical and about $67\times$ faster.
\end{remark}

\subsection{The memory term}

Each branch carries an exponentially decaying memory of the accumulated ridge correction:
\begin{equation}\label{eq:wo}
w_{j+1} = \lambda\, w_j + (1-\lambda)\,\hat{w},
\end{equation}
where $\hat{w}$ is a fresh estimate at the current point and λ is the measured autocorrelation of the residuals at the checkpoint spacing.

\begin{remark}[A term that cancelled itself]
An earlier form updated w toward the branch-mean of the current offsets. Since offsets are recentred to mean zero after each pruning, this drove w to zero: measured, it decayed from -46.6 to zero within ten checkpoints, so the memory term had no effect at all. The current form \eqref{eq:wo} targets a fresh independent estimate instead.
\end{remark}

\subsection{Measuring λ}\label{sec:decay}

λ in \eqref{eq:wo} is the lag-CP autocorrelation of the residual series. Measuring it correctly turned out to require care.

The natural implementation --- take a sample of the checkpoints, then correlate adjacent sampled values --- is wrong. After subsampling m points from a range of M checkpoints, adjacent samples are M/m checkpoints apart, not one. At $N=5\times10^{7}$ with $m=3000$ this measures a lag of about $\num{16680}$ instead of 40.

We verified the diagnosis on a synthetic AR(1) series with known correlation 0.85:

\begin{center}

```
\begin{tabular}{lrr}
\toprule
Method & Measured $\lambda$ & Error\\
\midrule
Subsample then correlate & $-0.0355$ & $0.886$\\
Blocks of consecutive points & $+0.849$ & $0.001$\\
\bottomrule
\end{tabular}
\end{center}

The correct method takes short blocks of \emph{consecutive} checkpoints,
distributed across the range. This preserves the true lag while still
sampling the whole range.

The same correction applies to the $\pi$ side, where it also removed a
substantial cost: the previous implementation walked every checkpoint
from the first prime to $N$.

\begin{center}
\begin{tabular}{lrr}
\toprule
Method ($N=5\times10^{6}$) & $\lambda_\pi$ & Time\\
\midrule
Full walk (\num{1000138} points) & $0.99846695$ & $3.67$\,s\\
Blocks (\num{12000} points) & $0.99849368$ & $0.04$\,s\\
\bottomrule
\end{tabular}
\end{center}

Agreement to $2.7\times10^{-5}$, $84\times$ faster.

\section{The anomaly mechanism}\label{sec:anomaly}

\subsection{Description}

Each branch runs a four-phase cycle intended to widen the cloud when the
model appears to be under stress:

\begin{center}
\begin{tabular}{cll}
\toprule
Phase & Trigger / duration & Action\\
\midrule
0 & $|w|/(\Cemp\ln^2 n) \ge 0.30$ & enter phase 1\\
1 & 2 checkpoints & add candidates at $\pm|w|$\\
2 & 1 checkpoint & transit\\
3 & 2 checkpoints & add candidates at $\mp 0.4|w|$, then return to 0\\
```

```
\bottomrule
\end{tabular}
\end{center}

\subsection{Measured effect: negligible on the output}

We measured its effect by disabling it (setting the trigger threshold to
$10^{9}$) and re-running:

\begin{center}
\begin{tabular}{lrrr}
\toprule
& Width & Contained & Min.\ margin\\
\midrule
Enabled & \num{6911} & $8/8$ & $26.7\%$\\
Disabled & \num{6901} & $8/8$ & $26.7\%$\\
\bottomrule
\end{tabular}
\end{center}

A difference of $0.14\%$ in width and none at all in containment.

\subsection{But real on the internal structure}

The mechanism is not inert. Measured on the cloud itself rather than the
interval:

\begin{center}
\begin{tabular}{lr}
\toprule
Quantity & Change when enabled\\
\midrule
Median of the candidate cloud & $0.9\%$ more accurate\\
Dispersion of the cloud & $2.2\%$ lower\\
Candidates surviving the $\pi$ filter & $19\%$ more\\
\bottomrule
\end{tabular}
\end{center}

\begin{observation}\label{obs:extremes}
The internal improvement does not reach the interval because the interval
is determined by the two extreme candidates alone. Any change to the
density of the cloud between those extremes is invisible in the output.
\end{observation}

Observation \ref{obs:extremes} is a general property of the extraction
rule, and it recurs in \S\ref{sec:pctl} and \S\ref{sec:pA}: several
```

mechanisms that measurably improve the cloud leave the interval unchanged, for this reason.

\subsection{Why it was retained}

Removing it yields a 0.14% narrower interval --- within measurement noise, and not worth a change to a component whose interaction with the rest of the branching is not fully characterised. It is retained, with a switch to disable it for measurement.

\section{Operating modes}\label{sec:modes}

The four modes differ \emph{only} in how $\Cemp$ is obtained. The containment range, the estimator, the branching, and the intersection are identical in all four.

\begin{center}
\begin{tabular}{cllc}
\toprule
Mode & $\Cemp$ source & Practical ceiling & Nature\\
\midrule
A & Full sieve, stored & $\sim2\times10^{9}$ (memory) & Measured\\
B & Sieve in chunks, discarded & $\sim2\times10^{11}$ (time) & Measured\\
C & 10^{5}-point sample via \texttt{primecount}
  & $\sim2\times10^{13}$ & Measured $+$ margin\\
F & Extrapolation from prior scales & Tool limit & Estimated\\
\bottomrule
\end{tabular}
\end{center}

\subsection{Mode A: full sieve}

The primes are generated by a segmented sieve and held in memory. Every $n \in [n_0,N]$ is inspected, so $\Cemp$ is exact for the training range.

Storage is the constraint. Stored as 64-bit integers, 10^{9} primes occupy 8\,GB.

\paragraph{Compressed storage.}
We store gaps rather than values. Every prime after 2 is odd, so gaps between consecutive primes are even, and $\mathrm{gap}/2$ fits in one byte: the largest known gap below 2.3×10^{10} is about 464, giving $\mathrm{gap}/2 = 232 < 255$. With anchor points at fixed stride, any slice is reconstructed by a cumulative sum from the nearest anchor.

Memory at $N=10^{9}$: 8\,GB $\to$ 1\,GB. Reconstructed slices are bit-identical to the original array.

```
\begin{remark}[An edge case worth recording]
The gap $2\to3$ equals $1$, which is odd --- $2$ being the only even
prime. Halving it corrupts the representation. The prime $2$ is therefore
stored separately and compression begins at $3$.
\end{remark}

\begin{remark}[Peak memory, not final memory]
An initial implementation generated all primes and then compressed. This
leaves the peak at $8N$ bytes, which is where the crash occurs;
compression after the peak does not help. Compression now happens per
sieve segment, so the peak is the segment size.
\end{remark}

\subsection{Mode B: streaming}

The sieve runs in chunks; $\Cemp$ is updated on each chunk as it is
produced, and the chunk is discarded. Every point is still inspected, so
coverage of the training range is complete --- but memory is the chunk
size, about $8$\,MB, independent of $N$.

\begin{observation}
Mode B was verified to produce $\Cemp$ \emph{identical to the last bit}
against mode A. It is therefore mode A without the memory ceiling, and is
the preferred choice wherever sieving time permits.
\end{observation}

Two passes are required: one to collect the calibration grid for fitting
the coefficients, one to compute $\Cemp$ using them.

\subsection{Mode C: sampling via \texttt{primecount}}

Above the sieving ceiling, $\pn$ is obtained from \texttt{primecount} at
$10^{5}$ points distributed across $[n_0,N]$. No sieve, no storage.

Sampling loses the guarantee: a sample can miss the extremum. Measured
across five scales from $2\times10^{6}$ to $2\times10^{7}$, the ratio of
the sampled $\Cemp$ to the full value was

\begin{center}
\begin{tabular}{lr}
\toprule
Statistic & Value\\
\midrule
Minimum & $0.9532$\\
Mean & $0.9779$\\
Standard deviation & $0.0131$\\
```

```
\bottomrule
\end{tabular}
\end{center}

Covering the worst observed shortfall requires $1/0.9532 = 1.049$;
covering mean $-\,3\sigma$ requires $1.065$. We use $1.10$, wider than
both, costing $10\%$ in interval width.

\begin{remark}[A failed attempt to make this self-calibrating]
We tried computing the shortfall from the current run, using the
saturation ratio $\Cemp(50\text{K})/\Cemp(100\text{K})$ as an indicator.
Its correlation with the actual shortfall was $r=-0.32$ --- weak and of
the wrong sign. The margin is therefore estimated from prior
measurements, not derived from the run at hand.
\end{remark}

\subsection{Mode F: extrapolation}

Above the sampling ceiling, $\Cemp$ is not measured at all. It is
extrapolated from a stored series of prior calibrations using a local
exponent from the last two scales,
\begin{equation}\label{eq:extrap}
\Cemp(N) \approx \Cemp(N_{-1})
\left(\frac{N}{N_{-1}}\right)^{k},\qquad
k = \frac{\ln\left(\Cemp(N_{-1})/\Cemp(N_{-2})\right)}
         {\ln\left(N_{-1}/N_{-2}\right)},
\end{equation}
multiplied by a safety margin (\S\ref{sec:margin}).

The local exponent was chosen over a global fit by measurement: mean
prediction error $16.1\%$ against $22.9\%$.

\begin{remark}[Mode C was dead code]
In an earlier version mode C required the sieved array for its
coefficient grid. But mode C's range begins \emph{above} the sieving
ceiling, so whenever it would have been selected the array was absent and
control fell through to mode F. Mode C was never chosen. It now builds
its grid from \texttt{primecount} exactly as mode F does; the only
difference between them is that C \emph{measures} $\Cemp$ by sampling
while F \emph{predicts} it.
\end{remark}

\subsection{Grid size is computed, not fixed}\label{sec:grid}

Modes C and F build their coefficient grid by calling
\texttt{nth\_prime}, whose cost grows steeply:
```

```
\begin{center}
\begin{tabular}{lr}
\toprule
$n$ & Cost per call\\
\midrule
$10^{9}$ & $5.8$\,ms\\
$10^{12}$ & $151$\,ms\\
$10^{13}$ & $622$\,ms\\
$2^{50}$ & $\approx 22$\,s\\
\bottomrule
\end{tabular}
\end{center}

A fixed grid of $\num{20000}$ points, chosen without reference to this
cost, implies $55$ hours at $2^{50}$. The grid size is therefore computed
at run time from a measured cost of one call and a stated time budget.

The floor of the grid size is justified by measurement. The coefficients
are mean-type statistics and saturate quickly:

\begin{center}
\begin{tabular}{rr}
\toprule
Grid points & Out-of-training estimation error\\
\midrule
50 & \num{1365}\\
100 & \num{1036}\\
200 & \num{864}\\
500 & \num{880}\\
2{,}000 & \num{974}\\
200{,}000 & \num{1039}\\
\bottomrule
\end{tabular}
\end{center}

From $100$ points upward all values lie in a single band and the
variation is not directional. Only at $50$ does the estimate degrade
clearly. Note that this concerns the \emph{coefficients} only; $\Cemp$,
being a maximum, does require many points, which is why modes C and F
handle it separately.



\section{Results}\label{sec:results}

\subsection{Containment across scales}
```

Every target is at ratio exactly 2.00. True values were obtained from \texttt{primecount}, and independently from a segmented sieve at $n=2\times10^{9}$ as a cross-check on the tool itself.

\begin{center}
\begin{tabular}{llccrr}
\toprule
N & Target $2N$ & Mode & Contained & Edge margin & vs.\ Axler\\
\midrule
5×10^{7} & 10^{8} & A & yes & --- & ---\\
5×10^{8} & 10^{9} & A & yes & --- & ---\\
10^{9} & 2×10^{9} & A & yes & --- & ---\\
5×10^{11} & 10^{12} & C & yes & 41.7% & $116\times$\\
2^{40} & 2^{41} & F & yes & 36.4% & $107\times$\\
2^{50} & 2^{51} & F & yes & 47.5% & $2{,}387\times$\\
2×10^{15} & 4×10^{15} & F & yes & 34.7% & $2{,}966\times$\\
3×10^{15} & 6×10^{15} & F & yes & 34.9% & $3{,}608\times$\\
\bottomrule
\end{tabular}
\end{center}

\begin{definition}
The \emph{edge margin} is
$\min\left(\pn-x_1,\ x_2-\pn\right)/(x_2-x_1)$. A value near 50% means the true value lies near the centre of the interval; a value near 0 means it lies near an endpoint, and the containment was close to failing.
\end{definition}

Two features of the table deserve comment.

\paragraph{The ratio to Axler grows with scale.}
From $116\times$ at 10^{12} to $3{,}608\times$ at 6×10^{15}. This is not an improvement in the model but a property of the comparison: by \eqref{eq:axler-upper} the Axler width grows faster than the calibrated interval does.

\paragraph{The edge margins are comfortable and stable.}
They range from 34.7% to 47.5% with no downward trend as scale increases. Had the calibration been degrading with scale, one would expect these to shrink toward zero. They do not. This is evidence, though not proof, that the extrapolation in mode F is not breaking down over the range tested.

\subsection{Comparison with exponentially tight effective bounds}\label{sec:visser}

Visser's bounds \eqref{eq:visser} \cite{visser2025} share a structural feature with the present construction: both take an inverse counting function as the base, and both localise $\pn$ by bracketing its argument rather than by expanding a series. The difference lies in what supplies the width.

Visser's width comes from a \emph{proven} effective bound on $|\pi(x)-\li(x)|$ of the form $a\,x(\ln x)^b e^{-c\sqrt{\ln x}}$. Ours comes from a \emph{measured} maximum deviation. The consequence is the expected one: his interval is unconditional and ours is not.

\begin{remark}[An omission we should state]
We have not made a numerical comparison. Doing so properly requires fixing the constants (a,b,c) from a specific effective estimate and evaluating both intervals at the same targets; we did not do this. We record it as an omission rather than present a comparison we have not performed.

What can be said without computation is that the two answer different questions. His bound holds; ours is calibrated to observed behaviour. A reader who needs a guarantee should use \eqref{eq:visser}.
\end{remark}

A natural question, which we leave open, is whether the empirical calibration described here could be applied with \eqref{eq:visser} as the base in place of R^{-1}, and whether that would narrow the result further.

\subsection{Exhaustive verification at small scales}

Sampling a few hundred targets can hide rare failures (\S\ref{sec:sampling}). At scales where it is feasible we therefore tested \emph{every} integer in the range:

\begin{center}
\begin{tabular}{lrrr}
\toprule
Range & Targets & Containment & Mean width\\
\midrule
1.28--2.56×10^{6} & \num{1280000} & 100.0000% & \num{5588.9}\\
2.56--5.12×10^{6} & \num{2560000} & 100.0000% & \num{8105.2}\\
\bottomrule
\end{tabular}
\end{center}

$\num{3840000}$ consecutive targets, no failure, and no case where the intersection search failed to find a valid centre.

\begin{remark}[These runs predate three fixes]
They were performed with an implementation containing the Axler constant error, the $\Cemp$ cutoff error, and the corrupted zeta table (\S\ref{sec:defects}). The intervals reported there are therefore inflated by a known factor. The \emph{containment rate} is unaffected: a wider interval still contains.
\end{remark}

\subsection{Sampled versus exhaustive targets}\label{sec:sampling}

The distinction matters more than it might appear. In one configuration during development, an exhaustive scan of \num{18005} targets revealed five containment failures. Random samples of 400 targets --- the size used for most comparative measurements --- had not detected any of them.

A failure rate of $5/\num{18005} \approx 2.8\times10^{-4}$ gives a probability of about 11% that a sample of 400 contains at least one failure. Absence from such a sample is therefore weak evidence.

All containment rates in this paper are from exhaustive scans, or from the complete set of targets at that scale.

\subsection{Direct jump versus full walk}

The model can either walk from N to the target step by step, or jump directly, settling the branch state over a fixed number of steps. The walk is the more conservative procedure; the jump is what makes large scales feasible.

\begin{center}
\begin{tabular}{lrrr}
\toprule
& Targets & Contained & Mean width\\
\midrule
Full walk & \num{6000} & $6000/6000$ & 947.2\\
Direct jump & 15 & $15/15$ & 916.8\\
\bottomrule
\end{tabular}
\end{center}

Ratio 0.968: the jump is 3.2% \emph{narrower}, with equivalent containment. We had expected the walk to be at least as tight, and it is not; we do not have an explanation for the direction of the difference.

\begin{remark}
This comparison is at $N=\num{25000}$, small enough that the walk is

feasible. Whether the relation holds at larger scales is untested --- the walk at $N=1.28\times10^{6}$ took about four hours.
\end{remark}

\section{Negative results}\label{sec:negative}

This section records approaches that were tried and failed. We include them because several are natural first attempts that look promising until measured, and because two of them (\S\ref{sec:normalt} and \S\ref{sec:anomloc}) bear on the structure of the problem rather than on implementation.

\subsection{Alternative normalisations}\label{sec:normalt}

Since $\ln^2 n$ has no theoretical basis for this use (\S\ref{sec:normjust}), we tested alternatives. The natural criterion is uniformity: a good normalisation makes $|r(n)|/\text{norm}$ roughly constant across the range, so that $\Cemp$ is not dominated by one region.

Measured as the ratio of maximum to minimum across four bins of the training range (ideal 1.0):

\begin{center}
\begin{tabular}{lr}
\toprule
Normalisation & Dispersion\\
\midrule
$\ln^2 n$ & $1.57\times$\\
$n^{0.4}$ & $1.11\times$\\
$\sqrt{n}$ & $1.10\times$\\
$\sqrt{n}/\ln n$ & $1.08\times$\\
\bottomrule
\end{tabular}
\end{center}

$\sqrt{n}/\ln n$ is the most uniform, and it is also the form the Riemann hypothesis suggests for the scale of $\pi(x)$ fluctuations --- so it has better motivation than $\ln^2 n$ on both counts.

It performs worse. Measured on actual targets, it \emph{widened} the interval by a factor of 1.8 ($\num{13914}$ against $\num{7846}$).

\begin{observation}\label{obs:extrapolate}
Lower dispersion \emph{inside} the training range does not imply better behaviour \emph{outside} it. The alternatives are more uniform on $[1,N]$ precisely because they grow faster, and that same growth inflates

```
them on $(N,2N]$.
\end{observation}

Observation \ref{obs:extrapolate} is, we think, the most transferable
finding in this section. The obvious criterion for choosing a
normalisation --- fit quality on known data --- selects the wrong answer
for an extrapolation problem.

\begin{remark}[Results do not transfer between architectures]
On a modified variant of the procedure, with the branching removed and
the fixed envelope substituted for the intersection, $n^{0.4}$ appeared
\emph{superior}. We initially adopted it on that basis and had to reverse
the decision. A measurement made on a simplified version of a system does
not necessarily hold for the system.
\end{remark}

\subsection{Percentile extraction}\label{sec:pctl}

By Observation \ref{obs:extremes} the interval is set by the two extreme
candidates, so improvements to cloud density are invisible. An obvious
response is to extract the bounds from a percentile rather than the
extremes.

Measured over $18$ targets at $N=2\times10^{6}$:

\begin{center}
\begin{tabular}{lrrr}
\toprule
Extraction & Width & Contained & Min.\ margin\\
\midrule
min/max & \num{7365} & $18/18$ & $21.7\%$\\
$0.5\%$ percentile & \num{7032} & $18/18$ & $22.0\%$\\
$2\%$ percentile & \num{6578} & $18/18$ & $18.8\%$\\
$5\%$ percentile & \num{6003} & $18/18$ & $16.0\%$\\
\bottomrule
\end{tabular}
\end{center}

The narrowing is real. We did not adopt it, for a reason that is
principled rather than empirical: the trimmed candidates were generated
by the model as admissible. Discarding them converts the procedure from
``the interval covering everything the model admits'' to ``the interval
covering most of what the model admits'', which is a different and weaker
statement. The measured containment does not distinguish the two at $18$
targets.

The option remains in the implementation, disabled by default.
```

\subsection{Self-calibrating the sampling margin}

Described in \S\ref{sec:modes}: the saturation ratio $\Cemp(50\text{K})/\Cemp(100\text{K})$ was tested as an indicator of how much the sample was missing. Correlation with the actual shortfall: $r=-0.32$, weak and of the wrong sign.

\subsection{Probability-weighted branching}\label{sec:pA}

A natural extension of the anomaly mechanism is to weight the added candidates by an estimated anomaly probability p and amplitude A, adding candidates at $\pm pA$ rather than at fixed offsets.

\begin{center}
\begin{tabular}{lrrr}
\toprule
& Width & Median error & Surviving candidates\\
\midrule
No branching & \num{6461} & 677 & 427\\
$p=0.3$ & \num{6524} & 700 & \num{1268}\\
$p=0.6$ & \num{6529} & 741 & \num{1233}\\
\bottomrule
\end{tabular}
\end{center}

The cloud responds strongly --- surviving candidates roughly triple --- but neither the width nor the median error improves. The branches distribute symmetrically in both directions, so the median does not move and the extremes move outward slightly.

What the approach requires, and lacks, is a \emph{direction}: a reason to add candidates on one side rather than both. That reduces to predicting where the anomaly lies, which is the subject of the next subsection.

\subsection{Predicting anomaly locations}\label{sec:anomloc}

Define an anomaly as a point in the top 1% of $|r(n)|$. Two facts about anomalies are established by measurement (\S\ref{sec:clustering}): they cluster strongly, and residual persistence is higher inside anomalous regions.

Neither fact yields a predictor of \emph{where} the next anomaly falls in the unknown region. Six candidate signals were tested against the signed error:

\begin{center}

\begin{tabular}{lr}
\toprule
Signal & Correlation with error\\
\midrule
$|w|$, the accumulated ridge correction & -0.10\\
Zeta zeros 61--260 as additional features & -0.036\\
Disagreement between the $\pn$ and π models & $+0.001$\\
Number of candidates surviving the π filter & -0.013\\
Residual at a single anchor point & decays beyond 10^{5}\\
Average over a large window & ≈ 0\\
\bottomrule
\end{tabular}
\end{center}

The first entry is worth its own comment: $|w|$ correlates
\emph{negatively}. The mechanism fires when the model is performing
\emph{better} than usual, which is the opposite of the intended
behaviour.

\begin{observation}\label{obs:structural}
We suggest the obstruction is structural. The anomaly is, by
construction, the component the model does not capture. Any signal
computed from the model is a function of what the model does capture.
There is no reason to expect the latter to inform the former, and
measurement finds that it does not.
\end{observation}

\subsection{A measurement that was an artefact}

An early attempt to test whether a branch's state predicts the direction
of its error reported $r=+0.70$ for the offset o --- an encouraging
result that turned out to be arithmetic rather than informational.

Writing the error of branch j as
\begin{equation}
e_j = \left(\text{base} + o_j + s_j d\right) - \pn,
\end{equation}
and noting that base and $\pn$ are identical across branches at a given
n, we have $e_j = o_j + \text{const}$. The correlation of o with e
is therefore 1 by construction, and the measured 0.70 merely reflects
the contribution of the $s_j d$ term.

The correct test is whether the branch nearest the truth occupies a
predictable position in the cloud ordering. Measured across six scales,
its normalised rank was
\[
0.737,\ 0.316,\ 0.564,\ 0.608,\ 0.555,\ 0.515,

\]
mean 0.549, standard deviation 0.125 --- consistent with uniform. The nearest branch is in no predictable position.

\begin{remark}
We record this because the erroneous version was checked and believed before the arithmetic was examined. A correlation of 0.70 between two quantities one of which is defined in terms of the other is not evidence of anything.
\end{remark}

\subsection{Asymmetric envelopes}

If the residual distribution were skewed, an asymmetric envelope would be narrower at equal coverage. It is not:

\begin{center}
\begin{tabular}{rrrr}
\toprule
Percentile & Upper tail & Lower tail & Ratio\\
\midrule
99.0 & 5.149 & -5.091 & 0.989\\
99.9 & 6.758 & -6.851 & 1.014\\
99.99 & 8.547 & -7.740 & 0.906\\
\bottomrule
\end{tabular}
\end{center}

Symmetric to within measurement error at every level tested.

\subsection{Global envelope widening}

If the entire unknown region were treated as anomalous, the envelope would reach its maximum width, which might be argued for as a safety measure. Measured:

\begin{center}
\begin{tabular}{lr}
\toprule
Quantity & Change\\
\midrule
Interval width & $+25.6\%$\\
Edge margin & $+4.6\%$\\
\bottomrule
\end{tabular}
\end{center}

A 5.6-to-1 loss. The arithmetic is immediate: the edge margin is $\min(\pn-x_1, x_2-\pn)/(x_2-x_1)$, and symmetric widening adds a fixed amount to the numerator while adding twice that to the denominator.

\subsection{A term that contradicted its own purpose}\label{sec:cremoved}

The base estimator originally included a correction
\begin{equation}
c(n) = a\sqrt{n\ln n} + b,
\end{equation}
fitted on the training range, intended to centre the model on the known sample.

After the base was upgraded from $\li^{-1}$ to R^{-1}, we tested whether it still earned its place. Measured on \num{7997} out-of-training targets, with a full ridge refit in each configuration:

\begin{center}
\begin{tabular}{lrr}
\toprule
Region & With $c(n)$ & Without\\
\midrule
Inside training (1--5×10^{6}) & 504 & $\mathbf{498}$\\
First 5×10^{5} beyond & \num{1200} & $\mathbf{930}$\\
6--8×10^{6} & 911 & $\mathbf{873}$\\
9--11.5×10^{6} & \num{1150} & $\mathbf{1063}$\\
\bottomrule
\end{tabular}
\end{center}

Better without it in every region --- including inside the training range, which is where it was supposed to help. Overall improvement 8.15%, $p = 1\times10^{-47}$ (Wilcoxon signed-rank).

The explanation is specific to the upgrade. With $\li^{-1}$ the base carried a systematic bias of order \num{47000}, and a curve of the form $a\sqrt{n\ln n}+b$ tracked it. With R^{-1} the bias is of order \num{6000} and no longer systematic in that shape, so the same curve fits noise. Worse, it was fitted on a 70.4% mask of the points (outliers excluded), whereas c_0 in \eqref{eq:Fz} is computed over all of them --- so the two centring mechanisms were not merely redundant but inconsistent.

$c(n)$ was removed. This is the clearest instance in the project of a component that was correct when written and became harmful when something upstream improved.

\section{The safety margin}\label{sec:margin}

\subsection{Why a margin is needed at all}

In modes A and B, $\Cemp$ is measured over every point of the training range and no margin arises. In mode C it is measured on a sample, and the margin covers what the sample misses --- a quantity we could measure directly (\S\ref{sec:modes}).

Mode F is different. There $\Cemp$ is not measured at all but extrapolated by \eqref{eq:extrap}, and the margin must cover the extrapolation error. That error has no measurable ceiling, and this section documents seven attempts to find one.

\subsection{Seven approaches, seven failures}

\paragraph{1. A constant chosen from observed failures.}
The first version used 1.55 for small scales and 1.25 for large, chosen after seeing which values had sufficed. When the first genuinely new scale was measured (5×10^{11}), the extrapolation from the prior series undershot by a factor of 1.75 --- so 1.55 would not have covered it. A constant fitted to past failures does not anticipate the next one.

\paragraph{2. A saturation indicator.}
Using $\Cemp(K)/\Cemp(2K)$ within the current run as a proxy for how close the series is to its limit: $r = -0.32$ against the actual requirement, and of the wrong sign.

\paragraph{3. A function of extrapolation distance.}
The intuition is that a longer extrapolation needs a wider margin. Measured over 146 pairs drawn from two series, the correlation between $\log(\text{jump})$ and the ratio (predicted/actual) was $r=+0.49$ --- that is, longer jumps tended to \emph{overshoot}, the opposite of the intuition. Fitting a margin law to this would have widened the interval in exactly the cases where it was already too wide.

\paragraph{4. A function of the anchor scale.}
$r = -0.31$, weak, and not monotone: the largest required margins occurred in the middle of the range, not at either end.

\paragraph{5. Self-measurement from prior errors.}
Compute the margin from the worst undershoot observed so far within the series. This works once a large undershoot has occurred and fails before it: measured coverage $8/9$, the single failure being the first large

jump, where the learned margin stood at 1.10 because every prior error had been positive.

\paragraph{6. Convergence to a limit.}
If the ratio (predicted/actual) converged, its limit would give the margin. It does not: measured values range over 0.386 to 1.26 with no narrowing as the series lengthens.

\paragraph{7. A Gaussian quantile.}
Model $\log(\text{predicted}/\text{actual})$ as normal and set the margin to $\exp(z\sigma - \mu)$. This produces a number and a nominal confidence level, and both are misleading. The normality assumption is unverified, and there is reason to doubt it: $\Cemp$ is a maximum-value statistic, and such statistics follow heavier-tailed families. A stated ``99% confidence'' describes the formula, not the process.

We report a cross-region validation of this approach because it is the only one that came close. Deriving (μ,σ) from scales 10^{4}--2×10^{7} and testing on an independent series 2.5--6.2×10^{7} gave zero failures in nine cases at all three confidence levels tested, with the actually required margin there being 1.179 against a derived 1.385. That is genuine cross-region validation --- but only for extrapolation jumps of $2.5\times$ or less, which is the operating range and not the regime where the margin is most needed.

\subsection{What was adopted}

Given that no derivation survived, we use the empirically safest available quantity: the largest ratio observed in \emph{paired} measurements, where $\Cemp$ was both extrapolated and directly measured at the same scale.

\begin{center}
\begin{tabular}{lrrr}
\toprule
Scale & Measured & Extrapolated & Required margin\\
\midrule
2×10^{6} & 8.009 & 10.309 & 0.777\\
4×10^{6} & 9.696 & 12.600 & 0.770\\
8×10^{6} & 14.313 & 12.001 & 1.193\\
1.2×10^{7} & 15.237 & 15.461 & 0.986\\
2×10^{7} & 19.224 & 18.705 & 1.028\\
2^{40} & 3681.800 & 2102.565 & $\mathbf{1.751}$\\
\bottomrule
\end{tabular}
\end{center}

The margin is 1.751, with no manual inflation. In four of the six cases the extrapolation already overshot, so no margin was required there.

\begin{remark}[An inflation that was removed]
An intermediate version used 1.85 --- the measured 1.751 plus a 6% increment added on the grounds that the measured value looked close to the edge. That increment had no source and has been removed. The margin is now exactly the worst measured value, and it is computed from the table above at run time, so adding a new paired measurement updates it automatically.
\end{remark}

\subsection{The weakness we cannot resolve}

Five of the six paired measurements lie between 2×10^{6} and 2×10^{7}. The sixth, at 2^{40}, is the worst of them --- and it is the only one at a large scale.

Two readings are consistent with the data: that the requirement is roughly constant and 2^{40} was an unlucky draw, or that it grows with scale and 2^{40} is the beginning of a trend. Distinguishing them requires paired measurements at 10^{9}, 10^{11}, 10^{12}, each of which costs hours. We flag this as the most consequential open item (\S\ref{sec:open}).

\section{Ancillary measurements}\label{sec:clustering}

The following were obtained while investigating the residual structure. They are not required by the construction, and we report them for their independent interest.

\subsection{Anomalies cluster}

Defining an anomaly as a point in the top 1% of $|r(n)|$:

\begin{center}
\begin{tabular}{lr}
\toprule
Quantity & Value\\
\midrule
Anomalous points adjacent to another anomaly & 93.6%\\
Expected under random assignment & 1%\\
$P(\text{anomaly at } n+d \mid \text{anomaly at } n)$, $d=10^{3}$ & 0.838\\
Baseline rate & 0.010\\
Ratio & $84\times$\\
\bottomrule

```
\end{tabular}
\end{center}

The clustering is strong and unambiguous. It is also, by
\S\ref{sec:anomloc}, not usable for prediction: knowing that anomalies
cluster does not locate the next cluster.

\subsection{Persistence: the Hurst exponent}

Rescaled-range analysis of the residual series gives $H = 0.348$.

Since $H < 0.5$, the series is mean-reverting rather than persistent: a
large positive residual is more likely than chance to be followed by a
negative one. This is consistent with the residual being an oscillation
around a correct trend, and inconsistent with any drift.

\begin{remark}
This result also closed a design question. An earlier version of the
anomaly mechanism included a ``rebound'' phase, on the assumption that a
large residual would be followed by one of opposite sign at a
characteristic distance. Direct measurement over $10^{7}$ points found
that residuals \emph{persist} at short range (\S\ref{sec:rho}) and revert
only over long ones, with no characteristic rebound distance. The phase
was based on behaviour not present in the data.
\end{remark}

\subsection{Short-range autocorrelation}\label{sec:rho}

\begin{center}
\begin{tabular}{lr}
\toprule
Distance $d$ & $\rho(d)$\\
\midrule
$10^{3}$ & $0.995$\\
$10^{4}$ & $0.964$\\
$10^{5}$ & $0.721$\\
$10^{6}$ & $-0.171$\\
$5\times10^{6}$ & $+0.001$\\
\bottomrule
\end{tabular}
\end{center}

Strong at short range, gone by $10^{6}$. The negative value at $10^{6}$
is within estimation noise (standard error about $0.058$ at the sample
sizes used).

\paragraph{Conditional on anomaly state.}
```

Persistence is higher inside anomalous regions:

\begin{center}
\begin{tabular}{lrr}
\toprule
Distance & Inside anomaly & Outside\\
\midrule
10^{4} & 0.954 & 0.858\\
3×10^{4} & 0.836 & 0.673\\
6×10^{4} & 0.41 & 0.00\\
\bottomrule
\end{tabular}
\end{center}

\subsection{The online correction}\label{sec:online}

Because $\pn$ is known exactly up to N, the residual at N is known for free. By \S\ref{sec:rho} it is informative about residuals a short distance beyond. The optimal linear predictor is
\begin{equation}\label{eq:online}
\hat{r}(N+d) = \rho(d)\, r(N),
\end{equation}
with ρ selected according to whether the anchor lies in an anomalous region.

Measured against the uncorrected estimate, 400 targets per distance:

\begin{center}
\begin{tabular}{lrrrl}
\toprule
d & Uncorrected & Corrected & Gain & p (Wilcoxon)\\
\midrule
10^{4} & \num{1095} & \num{1028} & 6.1% & 0.005\\
3×10^{4} & \num{1959} & \num{1759} & 10.2% & 5×10^{-4}\\
6×10^{4} & \num{2845} & \num{2424} & 14.8% & 4×10^{-13}\\
10^{5} & \num{3350} & \num{2729} & 18.5% & 4×10^{-15}\\
\bottomrule
\end{tabular}
\end{center}

Beyond 10^{5} the gain vanishes, and \eqref{eq:online} returns the uncorrected estimate automatically since $\rho\to0$.

\begin{remark}[This is the correct form of an idea that first failed]
An earlier attempt added an independent anomaly term pA \emph{on top of} \eqref{eq:online}, and degraded the estimate by 57.9% at $d=2000$. The reason is that \eqref{eq:online} is already the optimal

linear predictor given $r(N)$; anything added to it double-counts. The correct use of the anomaly information is not to add a term but to select a different ρ, which is what the table above does.
\end{remark}

\begin{remark}[Guarding against spurious negative ρ]
The first implementation interpolated ρ across all measured distances, including a spurious $\rho=-0.331$ at $d=3\times10^{5}$ that was noise. This produced an 8.6% \emph{degradation} at $d=2\times10^{5}$. The current version truncates at the first distance where $|\rho|$ falls below a trust threshold and zeroes everything beyond, so the correction switches off cleanly instead of doing harm.
\end{remark}

\subsection{A range law}

Combining \S\ref{sec:rho} with the residual magnitudes suggests that the distance over which an observed residual remains informative scales as
\begin{equation}
d_{\max} \approx \left(\frac{r}{\ln x}\right)^{2}.
\end{equation}
This is an empirical fit, not a derivation. It is consistent with the measured decay of ρ across the scales tested.

\section{Implementation defects}\label{sec:defects}

Several defects produced effects large enough to invalidate conclusions while leaving the containment rate intact --- that is, silently. We record them in full, since the pattern is instructive.

\begin{center}
\begin{longtable}{p{5.4cm}p{7.6cm}}
\toprule
Defect & Measured impact\\
\midrule
\endhead
Axler upper constant set to 0 instead of 10.667
  & Interval $13.7\times$ wider than necessary at $n=10^{7}$\\
$\Cemp$ cutoff at $n\ge10$ (\S\ref{sec:cutoff})
  & $\Cemp$ inflated $8.5\times$\\
Corrupted zeta-zero table
  & One entry (109.18895) is not a zero; one genuine zero (111.029536) absent\\
Autocorrelation measured after subsampling (\S\ref{sec:decay})
  & Measured -0.036 where the truth was 0.849\\
π-side decay inconsistent with $\pn$-side
  & Traversed 10^{7} points to obtain a value that $12{,}000$ points

```
    give to $2.7\times10^{-5}$\\
Memory term cancelling itself (\S\ref{sec:branching})
  & Decayed to zero within ten checkpoints; no effect\\
$c(n)$ retained after the base upgrade (\S\ref{sec:cremoved})
  & Error $+8.15\%$, $p=1\times10^{-47}$\\
Convex hull allocating $O(N)$ float arrays
  & Out-of-memory at $N=5\times10^{8}$\\
Gap $2\to3$ is odd
  & Corrupted the compressed prime representation\\
Misaligned calibration series
  & $\Cemp$ for the $\pi$ side $1{,}278\times$ too small\\
\bottomrule
\end{longtable}
\end{center}

\subsection{The silent failure}

The last entry merits detail, since it is the clearest example of the
pattern.

Two parallel series --- calibration values for the $\pn$ side and for the
$\pi$ side --- were filtered by different conditions and then truncated
to a common length. The truncation dropped the last element of one
series, so every value in it was paired with a scale $\num{24000}$ times
smaller than its own. The extrapolated exponent collapsed, and $\Cemp$
for the $\pi$ side came out $1{,}278\times$ too small.

The consequence: the $\pi$ cloud narrowed to near-nothing, the
intersection \eqref{eq:intersect} became vacuous, and the reported
interval was determined entirely by the Axler clipping
\eqref{eq:clip}.

\emph{Containment still succeeded} --- the Axler bound being
unconditional --- so the run appeared healthy. The only symptom was that
the reported ratio to Axler had fallen to $1.0$, meaning the calibrated
component was contributing nothing.

\begin{remark}
Three runs totalling about nine hours were performed before this was
noticed. The lesson is specific: when a system has a proven fallback, the
fallback masks failure in the component under test. A ratio of $1.0$
against the fallback should be treated as an error condition, not as a
result.
\end{remark}

\section{Open questions}\label{sec:open}
```

\subsection{Does the required margin grow with scale?}

The most consequential unresolved question (\S\ref{sec:margin}). Five of six paired measurements are at 2--20×10^{6}; the single large-scale point, at 2^{40}, is also the worst. Resolving this requires paired measurements at 10^{9}, 10^{11} and 10^{12} --- each a run of hours, but entirely feasible.

If the requirement grows, mode F at large scales is under-protected and the margin should be a function of scale rather than a constant. If it does not, the current constant is adequate and the 2^{40} point was a draw from the tail.

\subsection{Is the anomaly location predictable at all?}

Six approaches failed (\S\ref{sec:anomloc}), and we offered a structural reason (Observation \ref{obs:structural}). But that reason, if correct, implies something stronger than we have shown: that \emph{no} quantity derived from the model can predict its own failures. That is a claim about the class of estimators, and we have tested six members of it.

An estimator built on information the current model does not use --- the higher zeros in a different parameterisation, or local structure in the prime gaps themselves --- is not excluded by our measurements.

\subsection{Why is the jump narrower than the walk?}

The direct jump gives intervals 3.2% narrower than the full walk, with equivalent containment. We expected the reverse. The comparison was made at a single small scale where the walk is feasible, and we have no explanation for the direction of the difference.

\subsection{Would $\pi(x)$-based grids raise the ceiling?}

All modes are bounded by the \texttt{nth_prime} limit of the available tool, 2.16×10^{17}, because the calibration grid is built by inverse queries. The forward function $\pi(x)$ reaches considerably further. Rebuilding the grid in terms of $\pi(x)$ and inverting locally would raise the ceiling, at the cost of an extra inversion per grid point. We have not implemented or costed this.

\subsection{Is $\ln^2 n$ the right normalisation?}

Section \ref{sec:normalt} shows it outperforms the alternatives tested, and \S\ref{sec:normjust} shows it has no theoretical claim to the role. Both can be true. A normalisation derived from the residual structure rather than borrowed from the gap literature might do better; we did not

```
find one.

\section{Conclusion}

The procedure yields an empirically calibrated interval for $\pn$,
$107$--$3{,}608$ times narrower than the tightest proven bounds, with a
measured containment rate of $100\%$ over eight scales up to
$6\times10^{15}$ and over $3.84$ million exhaustively tested targets at
smaller scales.

It is not a theorem, not a faster algorithm for computing $\pn$, and not
a guarantee. Its contribution is a direct measurement of how far analytic
estimates deviate in practice, an interval calibrated to that
measurement, and a documented account of which approaches to tightening
it do and do not work.

The defensible form of the claim is: \emph{an empirical interval, narrower
by a measured factor than proven bounds, with a containment rate measured
over a stated number of targets, and no theoretical guarantee.}

\appendix

\section{Parameters and their basis}

\begin{center}
\begin{tabular}{llp{6.2cm}}
\toprule
Parameter & Value & Basis\\
\midrule
Zeta zeros $K$ & 60 & Measured; more degrades (\S\ref{sec:morezeros})\\
$\Cemp$ cutoff $n_0$ & \num{10000} & Measured; stable from here
  (\S\ref{sec:cutoff})\\
Normalisation & $\ln^2 n$ & Measured against alternatives
  (\S\ref{sec:normalt}); no theoretical basis (\S\ref{sec:normjust})\\
Envelope multiplier & $\pm2$ & Design choice, not optimised\\
Margin floor & $1.751$ & Computed from paired measurements
  (\S\ref{sec:margin})\\
Sampling margin (mode C) & $1.10$ & Covers worst measured shortfall
  plus safety (\S\ref{sec:modes})\\
Checkpoint spacing $\mathrm{CP}_{pn}$, $\mathrm{CP}_{\pi}$ & 40, 5
  & Design choice\\
Branch cap / retained ($\pn$) & 128 / 64 & Design choice\\
Branch cap / retained ($\pi$) & 1024 / 512 & Design choice\\
Settling steps & 800 & Design choice\\
Boundary seeds & 5 & Sufficient for stability (\S\ref{sec:seed})\\
Grid time budget & 900\,s & User-set; grid size derived
  (\S\ref{sec:grid})\\
```

```
Axler upper constant & $10.667$ & \cite{axler2019}, Thm 1.1\\
Axler lower constant & $11.508$ & \cite{axler2019}, Thm 1.2\\
\bottomrule
\end{tabular}
\end{center}

Parameters marked ``design choice'' were set at the outset and not
subsequently optimised. We do not claim they are optimal; we claim only
that the results reported were obtained with these values.

\section{Code and data availability}

The implementation, eight experiment scripts with embedded true values
for automatic verification, and an independent benchmarking tool are
available at \url{[repository URL]}.

Each experiment script is self-contained and reproduces one row of the
table in \S\ref{sec:results}. The benchmarking tool measures the limits
and per-call costs of \texttt{primecount} on the reader's own hardware,
and verifies every true value used in this paper.


\begin{thebibliography}{9}

\bibitem{axler2019}
C.~Axler,
\newblock New estimates for the $n$-th prime number,
\newblock \emph{Journal of Integer Sequences} \textbf{22} (2019),
  Article 19.4.2. arXiv:1706.03651.

\bibitem{cramer1936}
H.~Cram\'er,
\newblock On the order of magnitude of the difference between consecutive
  prime numbers,
\newblock \emph{Acta Arithmetica} \textbf{2} (1936), 23--46.

\bibitem{dr1996}
M.~Del\'eglise and J.~Rivat,
\newblock Computing $\pi(x)$: the Meissel--Lehmer--Lagarias--Miller--Odlyzko
  method,
\newblock \emph{Mathematics of Computation} \textbf{65} (1996), 235--245.

\bibitem{dusart2018}
P.~Dusart,
\newblock Explicit estimates of some functions over primes,
\newblock \emph{Ramanujan Journal} \textbf{45} (2018), 227--251.
```

```
\bibitem{gourdon2001}
X.~Gourdon,
\newblock Computation of $\pi(x)$: improvements to the Meissel, Lehmer,
  Lagarias, Miller, Odlyzko, Del\'eglise and Rivat method,
\newblock unpublished (2001).

\bibitem{granville1995}
A.~Granville,
\newblock Harald Cram\'er and the distribution of prime numbers,
\newblock \emph{Scandinavian Actuarial Journal} \textbf{1} (1995), 12--28.

\bibitem{lmo1985}
J.~C. Lagarias, V.~S. Miller and A.~M. Odlyzko,
\newblock Computing $\pi(x)$: the Meissel--Del\'eglise--Lehmer method,
\newblock \emph{Mathematics of Computation} \textbf{44} (1985), 537--560.

\bibitem{rosser1962}
J.~B. Rosser and L.~Schoenfeld,
\newblock Approximate formulas for some functions of prime numbers,
\newblock \emph{Illinois Journal of Mathematics} \textbf{6} (1962), 64--94.

\bibitem{visser2025}
M.~Visser,
\newblock The $n$th prime exponentially,
\newblock arXiv:2504.14458 (2025).

\bibitem{primecount}
K.~Walisch,
\newblock \texttt{primecount},
\newblock \url{https://github.com/kimwalisch/primecount}.

\end{thebibliography}

\end{document}
```